Context:

- Fragmented academic data across HCMUS platforms (Portal for schedules, Moodle for assignments, etc) and a hectic schedule overwhelm first and second year students.
- A unified system utilizing natural language processing is required to resolve time management inefficiencies and consolidate scheduling operations within a 10-week project arc.

Stakeholder:

- **User:** HCMUS students.
- **Owner:** 6-member student development team (Ngô Vũ Khôi Nguyên, Huỳnh Nhật Duy, Nguyễn Tường Lâm, Đinh Quang Thịnh, Trương Anh Khôi, Phạm Phúc Nguyên).

Input:

1. **Initialization Input:** Timetable files (unstructured PDF, image for OCR, markdown, or raw text).
2. **Interactive Scheduling Input:** Natural language queries defining temporal constraints and task details.
3. **Qualification Input:** Priority classifications assigned by the user or inferred by AI.
4. **In-chat Input:** Explicit state confirmations (Accept, Reject, Reschedule).
5. **Sign in information:** Username/passwork, OAuth Token (Google, Github)

Output:

- **User-Facing Output:**

1. Visual calendar interface rendering occupied nodes and suggested available slots.
2. Conversational responses providing scheduling rationale.
3. Automated event notifications triggered prior to flagged events via notification.

- **System-Facing Output (IT Team):**
    1. Pipeline latency and profiling logs mapping processing time across parsing, search, and synthesis nodes.
    2. Infrastructure telemetry logging VRAM/RAM utilization and queue depth.
    3. AI error taxonomy logs categorizing failed parsing attempts to optimize prompts.
    4. Scheduler audit trails tracking background notification processes.

Constraint:

- **Correctness:** Zero overlap guarantee against fixed academic hard constraints.
- **Latency & UX:** End-to-end system response for natural language queries must execute within 2-4 seconds.
- **Data Privacy:** Strict isolation of personal schedule data; prohibited transmission to unsecured public APIs.
- **Operational Platform:** Local host execution or basic web deployment optimized for faculty evaluation.
- **Time:** 10-week project duration.

Assumption:

- The host hardware (i9-14900K, 64GB RAM, RTX 4090 24GB) processes inferences via a sequential request queue model (1-2 concurrent active inferences) rather than sustained parallel load to maintain optimal Time-To-First-Token (TTFT) throughput.
- Client network connectivity remains continuous.
- Users will input unstructured, ambiguous, and non-standardized phrasing (e.g., colloquialisms, abbreviations) rather than strictly formatted prompts.

Edge Case:

- **Domain-Specific:**
    1. **No Available Slot:** User requests a duration exceeding any continuous free block, necessitating task splitting or rescheduling lower-priority entities.
    2. **Temporal Ambiguity:** Vague natural language time references requiring relative date resolution.
    3. **Constraint Conflict:** User explicitly requests scheduling a task over a fixed hard constraint.
    4. **Session State Abandonment:** User initiates a new intent without explicitly confirming or rejecting previous system suggestions.
- **System/Input:** Non-diacritic Vietnamese, teencode, profanity, and unsupported special characters.

Success Criteria

- **Constraint Satisfaction:** 100% of suggested slots guarantee zero conflict with hard constraints; 100% temporal matching with requested task duration.

- **AI Pipeline Reliability:** >= 98% AI adherence to structured JSON schemas; Zero temporal hallucinations (100% of temporal facts align strictly with database parameters).
- **Performance:** Deterministic availability engine resolves 1-semester schedules in <= 200ms; End-to-end AI latency <= 4 seconds on the designated host hardware.
- **Fault Tolerance:** 100% of AI timeouts or failures execute a graceful fallback to a manual UI selection state in <= 1 second without application termination.